

– TECHNICAL DATA SHEET

NN-AluBond High Alumina Binder

High-Alumina Refractory Binder — Calcium Aluminate Cement (CAC)

Grades CA-50 / CA-70 / CA-80 • rapid-hardening hydraulic refractory binder to IS 15895



NN-AluBond — grey/buff calcium aluminate binder, supplied in 25 / 50 kg lined bags

Document No.	Revision	Issue Date	Product Code
NNPL/TDS/PROD-09	1.0	2 July 2026	NN-AluBond

01 PRODUCT DESCRIPTION

NN-AluBond is a high-alumina refractory binder — a calcium aluminate cement (CAC) in which hydraulic calcium aluminates, chiefly monocalcium aluminate ($\text{CaO} \cdot \text{Al}_2\text{O}_3$), are the primary strength-forming phase. It is a rapid-hardening hydraulic binder that develops high early strength and offers high refractoriness, resistance to abrasion, and resistance to acidic, sulphate and biogenic-sulphide attack.

It is used principally as the binder in refractory castables, mortars and gunning mixes for furnace, kiln and ladle linings, and in specialty rapid-strength and corrosion-resistant construction concretes. Supplied in grades by alumina content — CA-50, CA-70 and CA-80. Conforms to IS 15895 (high alumina refractory cement); the structural grade also aligns with IS 6452.

02 USES & APPLICATIONS

- **Refractory** — Binder for refractory castables, mortars and gunning mixes (furnace, kiln, ladle and boiler linings).
- **Monolithics** — Monolithic linings requiring high refractoriness and high early strength.
- **Rapid repair** — Rapid-strength and cold-weather construction concretes and emergency repairs.
- **Corrosion resistance** — Sewer and effluent structures needing biogenic-sulphide / acid resistance.
- **Specialty** — Blended-cement and specialty formulations needing controlled setting/expansion.

03 GRADES (BY ALUMINA CONTENT)

Grade designation follows Al_2O_3 mass fraction; select grade by service temperature and castable design.

Grade	Alumina	Typical use
CA-50	Al_2O_3 50 – 60 %	General refractory & construction castables
CA-70	Al_2O_3 68 – 72 %	Low/ultra-low cement castables; high strength
CA-80	$\text{Al}_2\text{O}_3 \geq 80$ %	High-purity castables; highest refractoriness

04 CHEMICAL COMPOSITION (TYPICAL)

Typical industrial values consolidated from Indian manufacturer specifications; grade-dependent. A batch Certificate of Analysis accompanies each consignment.

Parameter	Typical
Alumina (Al_2O_3)	50 – 80 % (by grade)
Lime (CaO)	$\approx 26 - 36$ % (decreasing with higher grade)
Silica (SiO_2)	0.5 – 6.5 % (lower for high-purity grades)
Iron oxide (Fe_2O_3)	Low (grade-dependent)
Principal phase	Monocalcium aluminate ($\text{CaO} \cdot \text{Al}_2\text{O}_3$, 'CA')

05 PHYSICAL & PERFORMANCE PROPERTIES

Property	Value
Appearance	Fine powder, grey to buff / off-white (high-purity grades)
Fineness (Blaine)	$\geq 300 \text{ m}^2/\text{kg}$ (or ≤ 20 % residue on $45 \mu\text{m}$ sieve)
Initial setting time	≥ 30 minutes
Final setting time	≤ 6 hours
Strength development	Rapid — high early strength (approx. 24-hour strength near grade value)
Refractoriness	Up to $\sim 1770^\circ\text{C}$ (grade-dependent)
Nature	Hydraulic; hygroscopic (moisture-sensitive)

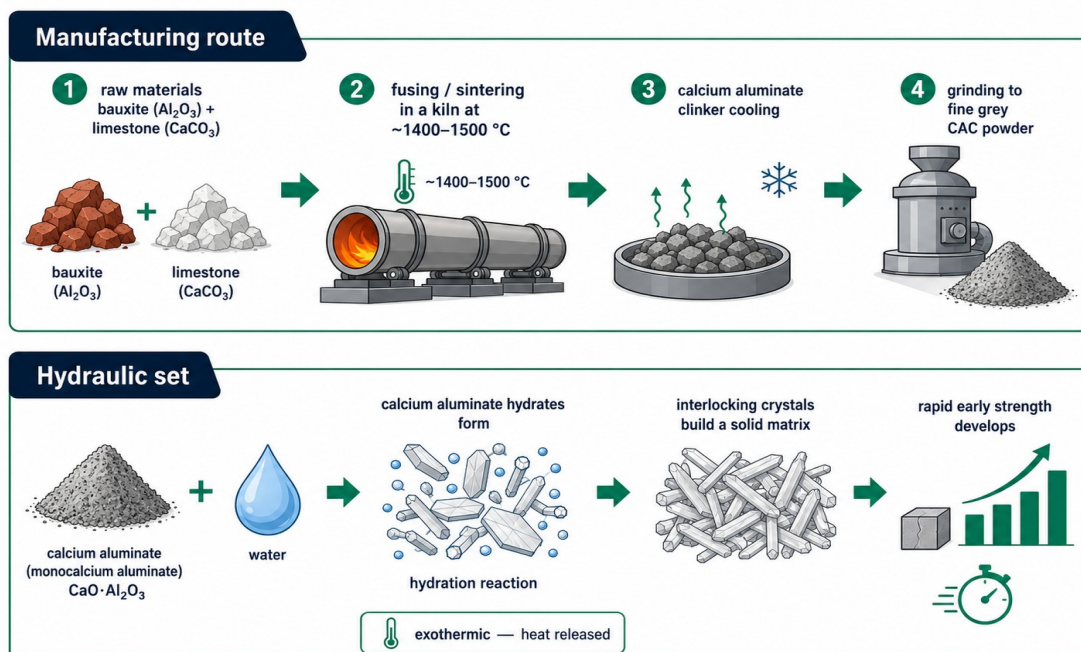
06 APPLICATION GUIDANCE



High-alumina binder used in refractory castables for furnace and kiln linings

- **Mix design** — Formulate castables with graded refractory aggregate; use the recommended binder proportion and water for the grade and design.
- **Mixing & placing** — Use clean water; mix thoroughly; place and vibrate promptly as the binder is rapid-setting.
- **Curing** — Cure in a moist condition for the first 24 hours at moderate temperature; avoid simultaneous high heat and humidity to limit conversion.
- **Dry-out** — Dry out and heat the lining gradually per the castable schedule before service.
- **Caution** — Do not blend with Portland cement or lime except in controlled, tested formulations (flash-set risk).

07 MANUFACTURING PROCESS



Calcium aluminate cement manufacturing route and hydraulic set

NN-AluBond is produced by fusing or sintering a controlled mix of an aluminous material (bauxite / calcined or fused alumina for high-purity grades) with a calcareous material (limestone) at high temperature. The resulting calcium aluminate clinker is cooled and ground to the finished cement. On mixing with water, the calcium aluminates hydrate and harden rapidly to develop early strength.

08 PACKAGING, STORAGE & SHELF LIFE

Item	Detail
Packaging	25 / 50 kg woven / paper bags with PE inner liner; jumbo bags / bulk to order
Storage	Store in a cool, dry, covered area on pallets; keep bags sealed — highly moisture-sensitive (pre-hydrates and loses activity)
Shelf life	6 months from manufacture in unopened, correctly stored, moisture-proof packaging
Handling	Use FIFO rotation; protect from humidity and rain; reseal opened bags promptly

09 PRECAUTIONS

- Irritant / alkaline** — Cementitious powder — alkaline when wet; may cause skin/eye irritation and burns on prolonged contact.
- Dust** — Avoid inhaling dust; wear gloves, goggles and a dust mask; wash after handling.

Refer to the NN-AluBond Safety Data Sheet for full physical, ecological, toxicological and handling information.

Disclaimer: The information and recommendations herein are given in good faith based on current knowledge. Performance depends on correct castable design, mixing, placing, curing and dry-out; users must establish suitability by trial. No warranty of merchantability or fitness for a particular purpose is implied. Specification values are indicative and grade-dependent; refer to the batch Certificate of Analysis and comply with all applicable standards (IS 15895 / IS 6452).